

COBORDISM OF SPIN MANIFOLDS WITH INVOLUTION

By V. GIAMBALVO

[Received 18 February 1975]

1. Introduction

Let $\Omega_*^{\text{Spin}}(\mathbf{Z}_2)$ denote the cobordism group of Spin manifolds with free orientation preserving involution. Specifically, define an equivalence relation (Spin- $\mathbf{Z}_2$ cobordism) on the set of pairs (T, M^n) where T is a free orientation preserving involution on the compact smooth n -dimensional manifold M^n as follows: (T_1, M_1) is Spin- $\mathbf{Z}_2$ cobordant to (T_2, M_2) if there is an $(n+1)$ -dimensional Spin manifold W with free orientation preserving involution T such that

- (1) W is a Spin cobordism between W_1 and W_2 ,
- (2) $T|_{M_i} = T_i$.

Let $\Omega_*^{\text{Spin}}(\mathbf{Z}_2)$ be the set of equivalence classes, and $\Omega_*^{\text{Spin}}(\mathbf{Z}_2) = \bigoplus \Omega_n^{\text{Spin}}(\mathbf{Z}_2)$. Then $\Omega_*^{\text{Spin}}(\mathbf{Z}_2)$ is a group under disjoint union.

Since the quotient space of a Spin manifold by a free orientation preserving involution does not necessarily have a Spin structure, one must look more deeply at the quotient space in order to compute $\Omega_*^{\text{Spin}}(\mathbf{Z}_2)$. In §2 a cobordism theory Ω_*^E , corresponding to manifolds whose second Stiefel-Whitney class is the product of two 1-dimensional classes is defined, and an onto homomorphism $\theta: \Omega_*^E \rightarrow \Omega_*^{\text{Spin}}(\mathbf{Z}_2)$ is constructed. Sections 3 and 4 are devoted to the computation of Ω_*^E via the Adams spectral sequence. These results are used in §5 to show that θ is a rational isomorphism, and an isomorphism in dimensions not congruent to 1 or 2 mod 8. It follows that $\Omega_*^{\text{Spin}}(\mathbf{Z}_2)$ has only 2 torsion. (The mod p analysis of $\Omega_*^{\text{Spin}}(\mathbf{Z}_2)$ for odd primes p is trivial, and has been omitted.) Finally some specific generators for Ω_*^E in low dimensions are given.

Since this paper will be concerned only with mod 2 behavior, all cohomology will be with $\mathbf{Z}_2$ coefficients, and $\mathcal{A}$ will denote the mod 2 Steenrod Algebra, and $\mathcal{A}_i$ the sub-Hopf Algebra of $\mathcal{A}$ generated by $Sq^0 = 1, Sq^1, Sq^2, \dots, Sq^{2^i}$.

The author wishes to thank R. Stong, who suggested the approach of this paper and outlined several of the proofs of §2. Acknowledgment is also due to the University of Connecticut Computer Center for providing a large amount of time which was instrumental in finding the model for the structure of $H^*(ME)$ and in the proof of Theorem 3.6.

2. Definition of Ω_*^E and its relation to $\Omega_*^{\text{Spin}}(\mathbb{Z}_2)$

Let $f: BSO \times BO_1 \times BO_1 \rightarrow K(\mathbb{Z}_2, 2)$ be the map inducing $f^*(i_2) = w_2 \otimes 1 \otimes 1 + 1 \otimes i_1 \otimes i_1$, where w_2 is the second Stiefel-Whitney class, and i_1, i_2 are the respective fundamental classes of BO_1 and $K(\mathbb{Z}_2, 2)$. Let $\bar{\pi}: E \rightarrow BSO \times BO_1 \times BO_1$ be the fibration induced from the path fibration over $K(\mathbb{Z}_2, 2)$ via f , and $\pi: E \rightarrow BSO$ the composition of $\bar{\pi}$ with projection onto the first factor. An E -structure on M is a pair α, β of classes in $H^1(M)$ together with a fiber homotopy class of liftings $\tilde{\nu}: M \rightarrow E$ of $\nu \times \beta \times \alpha: M \rightarrow BSO \times BO_1 \times BO_1$, where ν classifies the stable normal bundle of M . Two closed manifolds M_1, M_2 with E -structures $\tilde{\nu}_1, \tilde{\nu}_2$ are E -cobordant if there is a manifold W with E -structure $\tilde{\nu}_W$ such that

- (1) $\partial W = M_1 + (-M_2)$,
- (2) $\tilde{\nu}_W|_{M_i} = \tilde{\nu}_i \quad i = 1, 2$.

Denote by Ω_n^E the set of E -cobordism classes of closed n -manifolds with E -structure. Then Ω_n^E is a (B, f) cobordism theory in the sense of Stong [7] or Lashof [4], and from either of these one has the following result.

LEMMA 2.1. Ω_n^E is a group (under the operation of disjoint union). Moreover if ME denotes the Thom space of the bundle $\pi^1(\gamma)$ where γ is the universal vector bundle over BSO , then $\Omega_*^E = \bigoplus_n \Omega_n^E \approx \pi_*(ME)$.

THEOREM 2.2. There is an onto homomorphism $\theta: \Omega_*^E \rightarrow \Omega_*^{\text{Spin}}(\mathbb{Z}_2)$.

Proof. Let $a \in \Omega_n^E$ be represented by $(M, \tilde{\nu})$ and define $\alpha \in H^1(M)$ by $(\tilde{\pi}\tilde{\nu})^*(1 \otimes 1 \otimes i_1)$. Consider the diagram

$$\begin{array}{ccccc} \tilde{M} & \longrightarrow & \tilde{E} & \xrightarrow{\pi} & BSO \times BO_1 \times EO_1 \\ \rho \downarrow & & \bar{\rho} \downarrow & & \hat{\rho} \downarrow \\ M & \xrightarrow{\tilde{\nu}} & E & \xrightarrow{\bar{\pi}} & BSO \times BO_1 \times BO_1 \end{array}$$

where $\hat{\rho}$ is the canonical double cover, $\bar{\rho}$ and ρ are the induced double coverings, and hence $\tilde{M}$ is the double covering of M corresponding to $\alpha \in H^1(M)$. Since $\pi^*(w_2 \otimes 1 \otimes 1) = \bar{\pi}^* \bar{\rho}^*(w_2 \otimes 1 \otimes 1) = \bar{\rho}^* \bar{\pi}^*(w_2 \otimes 1 \otimes 1) = \bar{\rho}^*(1 \otimes i_1 \otimes i_1) = 0$, $\bar{\pi}$ lifts to a map $\tilde{E} \rightarrow B \text{Spin} \times BO_1 \times EO_1$ which is a homotopy equivalence. Hence $\tilde{M}$ has a Spin structure, and the covering transformation T gives a free involution on $\tilde{M}$ since M is oriented. Define $\theta(a) = [(T, \tilde{M})] \in \Omega_n^{\text{Spin}}(\mathbb{Z}_2)$. Clearly θ is additive. To see that θ is well defined, suppose $(M, \tilde{\nu})$ is E -cobordant to 0. Let $(W, \tilde{\nu}_W)$ be an E -cobordism. Let $\alpha' = \tilde{\nu}_W^* \bar{\pi}^*(1 \otimes i_1 \otimes i_1) \in H^1(W)$. Then the double cover of W corresponding to α' gives a Spin- $\mathbb{Z}_2$ cobordism of $\theta([M, \tilde{\nu}])$ with 0.

To show that θ is onto, let (T, M) represent an element of $\Omega_*^{\text{Spin}}(\mathbb{Z}_2)$. Consider the first few terms of the Gysin sequence for the $\mathbb{Z}_2$ bundle $\xi: M \rightarrow M/T$.

$$H^0(M/T) \xrightarrow{\cup \alpha} H^1(M/T) \xrightarrow{\zeta^*} H^1(M) \\ \xrightarrow{\psi} H^1(M/T) \xrightarrow{\cup \alpha} H^2(M/T) \xrightarrow{\zeta^*} H^2(M).$$

Here $\alpha \in H^1(M/T)$ is the characteristic class of the bundle. Since $0 = w_2(M) = \zeta^*(w_2(M/T))$, there is a class $\beta \in H^1(M/T)$ with $\beta \cup \alpha = w_2(M/T)$. Hence the map $\nu \times \beta \times \alpha: M/T \rightarrow BSO \times BO_1 \times BO_1$ lifts to E , and M/T has an E -structure. Note that the choice of β is not unique. However if $\beta' \cup \alpha = w_2(M/T)$, then $(\beta + \beta') \cup \alpha = 0$, so there is a $\gamma \in H^1(M)$ with $\psi(\gamma) = \beta + \beta'$. Once β is chosen, the lifting $\tilde{\nu}$ can be varied by an element of $[M/T, K(\mathbb{Z}_2, 1)] = H^1(M/T)$. Thus the choice of β can vary the Spin structure on the double cover of M/T by an element of $H^1(M)/\ker(\psi) = H^1(M)/\text{Im } \zeta^*$ and the choice of lifting can vary the Spin structure by an element of $\text{Im } (\zeta^*)$, so every Spin structure on M can be realized as $\theta([M/T, \tilde{\nu}])$ for some choice of E -structure on M/T . Thus θ is onto.

COROLLARY 2.3. *Let $x \in \Omega_n^{\text{Spin}}(\mathbb{Z}_2)$, $n > 3$. Then there is a representative (T, M) of x with M connected and $\pi_1(M)$ either 0 or $\mathbb{Z}_2$.*

Proof. Lashof [4] has shown that for any (B, f) cobordism theory, and any class $y \in \Omega_n^{(B, f)}$, one can perform surgery preserving the (B, f) cobordism class to obtain a representative $M, \tilde{\nu}: M \rightarrow B$ with $\tilde{\nu}_*: \pi_k(B) \rightarrow \pi_k(M)$ an isomorphism for $k < [n/2]$. One applies this construction to a representative $(M, \tilde{\nu})$ of $a \in \Omega_*^E$ with $\theta(a) = x$, to make M connected and to kill the kernel of $\tilde{\nu}_*: \pi_1(M) \rightarrow \pi_1(E)$. Since $\pi_1(E) \approx \mathbb{Z}_2 \oplus \mathbb{Z}_2$, $\pi_1(M)$ must be either 0, $\mathbb{Z}_2$, or $\mathbb{Z}_2 \oplus \mathbb{Z}_2$, and if it is $\mathbb{Z}_2 \oplus \mathbb{Z}_2$, $\tilde{\nu}_*$ is an isomorphism. Thus $\theta(M, \tilde{\nu})$ must have fundamental group 0 or $\mathbb{Z}_2$.

3. Computation of $H^*(ME)$ as an $\mathcal{A}$ -module

In this section the structure of the cohomology of the Thom space ME will be determined. This will be used in Section 4 to compute Ω_*^E . Let w_i denote both the i th Stiefel-Whitney class in $H^*(BSO)$ and its image in $H^*(E)$. In order to compute $H^*(ME)$ we will need the following version of a theorem of Margolis [6].

LEMMA 3.1. *Let M' and N' be $\mathcal{A}_1$ modules with $M = \mathcal{A} \otimes_{\mathcal{A}_1} M'$ and $N = \mathcal{A} \otimes_{\mathcal{A}_1} N'$. Let $Q_0 = Sq^1 \in \mathcal{A}$ and $Q_1 = Sq^2 Sq^1 + Sq^3$. Then Q_0 and Q_1 are differentials on M and N . If $f: M \rightarrow N$ is an $\mathcal{A}$ module map inducing an isomorphism $f_*: H(M, Q_i) \rightarrow H(N, Q_i)$, $i = 1, 2$, then there are free modules F_1 and F_2 and an $\mathcal{A}$ module isomorphism $g: M \oplus F_1 \rightarrow N \oplus F_2$ such that $g|_M = f$.*

To compute $H^*(ME)$ one needs to know $H^*(E)$. A standard spectral sequence argument gives the following lemma.

LEMMA 3.2. $H^*(E) \approx H^*(BSO) \otimes \mathbb{Z}_2[\alpha, \beta]/I(w_2 \oplus 1 + 1 \otimes \alpha\beta)$ where $\alpha = \bar{\pi}^*(1 \otimes 1 \otimes i_1)$, $\beta = \bar{\pi}^*(1 \otimes i_1 \otimes 1)$, and I denotes the ideal generated over $\mathcal{A}$. Furthermore w_i is decomposable, $i \geq 4$, if and only if $i = 2^j + 1$ for some j .

Thus $H^*(ME)$ is known as a $\mathbb{Z}_2$ -vector space, via the Thom isomorphism $\Phi: H^*(E) \rightarrow H^*(ME)$. Let $U \in H^0(ME)$ be the Thom class. Note that $Sq^1 U = 0$ and $Sq^2 U = \alpha\beta U = w_2 U$. Let $S = I(w_2 \otimes 1 + 1 \otimes \alpha\beta) \subset H^*(E)$. Then $H^*(ME) \approx \mathbb{Z}_2[\alpha, \beta] \otimes H^*(MSO)/S$ as $\mathbb{Z}_2$ -vector spaces. Since $H^*(MSO) \approx \bigoplus \mathcal{A} \oplus (\bigoplus \mathcal{A}/\mathcal{A}Sq^1)$, there is an $\mathcal{A}_1$ module X_1 with $H^*(MSO) \approx \mathcal{A} \otimes_{\mathcal{A}_1} X_1$, as $\mathcal{A}$ modules. Let X_2 be the $\mathcal{A}_1$ module $\mathbb{Z}_2[\alpha, \beta] \otimes X_1/S^1$, where S^1 is the submodule generated over $\mathcal{A}_1$ and X_1 by $w_2 \otimes 1 + 1 \otimes \alpha\beta$. Then $H^*(ME)$ is isomorphic to $\mathcal{A} \otimes_{\mathcal{A}_1} X_2$ as $\mathcal{A}$ modules. Thus to apply Lemma 1 we need a model for $H^*(ME)$ and $H(H^*(ME), Q_i)$, $i = 1, 2$.

LEMMA 3.3. As graded $\mathbb{Z}_2$ -modules $H(H^*(ME), Q_0)$ is isomorphic to $\mathbb{Z}_2[w_{2^i}^2, i \neq 2^j] \otimes \mathbb{Z}_2[u_{2^j}, j > 1]$, where $\deg(u_{2^j}) = 2^j$.

Proof. Since $Q_0 U = Sq^1 U = 0$, $H(H^*(ME), Q_0) \approx H(H^*(E), Q_0)$. The ring structure of $H^*(E)$ simplifies the proof. Let $B_1 \subset H^*(E)$ be the subalgebra generated by α, β , and $\{w_i, i \neq 2^j\}$. Since w_i is decomposable if $i \neq 2^j + 1$, $B_1 \approx \mathbb{Z}_2[\alpha, \beta] \otimes \mathbb{Z}_2[w_i, i \neq 2^j, 2^j + 1]$. Now $Q_0 w_{2^i} = w_{2^i+1}$, $Q_0(w_{2^i+1}) = 0$, $Q_0(\alpha^k) = k\alpha^{k+1}$ and $Q_0(\beta^k) = k\beta^{k+1}$. Hence $H(B_1, Q_0) \approx \mathbb{Z}_2[w_{2^i}^2, i \neq 2^j]$. Define $B_n \subset H^*(E)$, and $u_{2^n} \in H^{2^n}(E)$ inductively satisfying the following:

- (1) $Q_0(u_{2^n}) = 0$,
- (2) $u_{2^n} \equiv w_{2^n}$ modulo decomposables,
- (3) $B_{n+1} \approx B_n \otimes \mathbb{Z}_2[u_{2^{n+1}}]$,
- (4) $H(B_{n+1}, Q_0) \approx H(B_n, Q_0) \otimes \mathbb{Z}_2[u_{2^{n+1}}]$.

Let $u_4 = w_4 + \alpha^3\beta + \alpha\beta^3$. Since $w_5 = Sq^2 w_3 + w_3 w_2 = \alpha^4\beta + \alpha^2\beta^3 + \alpha^3\beta^2 + \alpha\beta^4$ we have $Q_0(u_4) = 0$ and (2) is clearly satisfied, so we can define B_3 by (3), and (4) holds.

To complete the induction we need only assume u_{2^j} and B_j defined for $j < n$ satisfying (1)–(4) and define u_{2^n} satisfying (1) and (2). Since w_{2^n+1} is decomposable $w_{2^n+1} \in B_{n-1}$. But $Q_0(w_{2^n+1}) = 0$. Since all the homology of B_{n-1} is in even dimensions, there is a class $d_{2^n} \in B_n$ with $Q_0(d_{2^n}) = w_{2^n+1}$. Since $w_{2^n} \notin B_{n-1}$, d_{2^n} is decomposable. Let $u_{2^n} = w_{2^n} + d_{2^n}$. Then $Q_0(u_{2^n}) = 0$ and 2 holds. Thus the induction is complete.

Since $\bigcup_n B_n = H^*(E)$, this completes the proof of Lemma 3.3.

LEMMA 3.4. *Let $z \in H^*(E)$ be the class $\alpha^2 + \alpha\beta + \beta^2$. Then as graded $\mathbf{Z}_2$ modules, $H(H^*(ME), Q_1) \approx z \cdot \mathbf{Z}_2[w_{2i}^2, i \neq 2^j - 2, i \geq 4] \otimes \mathbf{Z}_2[\tilde{u}_{2^j-2}, j \geq 3]$.*

Proof. This is similar to the proof of Lemma 3.3 above, but technically more complicated by the fact that $Q_1(U) = (\alpha^2\beta + \alpha\beta^2)U \neq 0$. One first computes $H(\Phi(Z_2(\alpha, \beta)), Q_1) = \mathbf{Z}_2$, with non zero class represented by z . This is merely tedious, but not difficult. One then proceeds inductively as in the proof of the preceding lemma to define classes $\tilde{u}_{2^j-2} \in H^*(E)$ with $Q_1(\tilde{u}_{2^j-2}) = 0$, and $u_{2^j-2} \equiv w_{2^j-2}$ modulo decomposables. The details are left to the reader.

We now construct the model for $H^*(ME)$.

Define $M'_0 = \mathcal{A}_1 \oplus s\mathcal{A}_1$, denoting the generators by x and y respectively (s is the algebraic suspension, increasing degree by 1). Let $M'_1 = M/S_1$, where S_1 is the submodule generated over $\mathcal{A}_1$ by Sq^1x and $Sq^3x + Sq^2y$. Let $M'_2 = M'/S_2$ where S_2 is the submodule generated by Sq^1y and $Sq^3Sq^3x + Sq^2Sq^3y$. Let $M_1 = \mathcal{A} \otimes_{\mathcal{A}_1} M'_1$ and $M_2 = \mathcal{A} \otimes_{\mathcal{A}_1} M'_2$.

LEMMA 3.5. *The Q_i homologies $i = 0, 1$ of M_j , $j = 1, 2$, are isomorphic as graded $\mathbf{Z}_2$ modules to the following:*

- (a) $H(M_1, Q_0) \approx \mathbf{Z}_2[\xi_1^4]$,
- (b) $H(M_1, Q_1) \approx s^2E(\xi_2^2, \xi_3^2, \dots)$,
- (c) $H(M_2, Q_0) \approx s\mathbf{Z}_2[\xi_1^4]$,
- (d) $H(M_2, Q_1) \approx s^3E(\xi_2^2, \xi_3^2, \dots)$,

where $\deg(\xi_i) = 2^i - 1$.

Proof. $H(M'_j, Q_i)$ is easy to compute and the spectral sequence from $H(M'_j, Q_i) \otimes H(\mathcal{A}/\mathcal{A}\mathcal{A}_1, Q_i)$ to $H(M_j, Q_i)$ collapses.

It is a consequence of Lemma 3.2 that $H^*(ME)$ is isomorphic as graded $\mathbf{Z}_2$ modules to $H^*(M\text{Spin}) \otimes \mathbf{Z}_2(\alpha, \beta)$. The structure of $H^*(M\text{Spin})$ as an $\mathcal{A}$ module will be used to construct the model for $H^*(ME)$.

Let V_0 be the graded $\mathbf{Z}_2$ -vector space with basis all finite sequences of integers $I = (i_1, \dots, i_k)$ such that $i_1 \leq i_2 \leq \dots \leq i_k$ and $i_1 > 1$. Let $n(I) = \sum_{j=1}^k i_j$. The degree of I will be $4n(I)$ if $n(I)$ is even, and $4n(I) - 2$ if n is odd. Let V_1 be the subspace of V_0 generated by all I with $n(I)$ even, and V_2 the subspace generated by all I with $n(I)$ odd. In [2] it was shown that $H^*(M\text{Spin}) \approx (\mathcal{A}/\mathcal{A}\mathcal{A}_1 \otimes V_1) \oplus (\mathcal{A}/\mathcal{A}Sq^3 \otimes V_2) \oplus F_1$ where F_1 is a free $\mathcal{A}$ module.

Let $M = M_1 \otimes V_1 \oplus M_2 \otimes sV_2$.

THEOREM 3.6. *There is an $\mathcal{A}$ module map $h: M \rightarrow H^*(ME)$ which induces an isomorphism on Q_i homology $i = 1, 2$.*

Proof. Let x_0 and x_1 denote the $\mathcal{A}$ generators of M_1 of degree 0 and 1 respectively, and y_0 and y_1 the $\mathcal{A}$ generators of M_2 . Any sequence $(i_1, \dots, i_k) \in V_0$ can be reordered as $(a_1, a_2, \dots, a_r, b_1, b_2, \dots, b_s, c_1, c_1, \dots, c_m, c_m)$ with each a_i even, $a_1 \leq a_2 \leq \dots \leq a_r$, each b_i and c_i odd, and $b_1 < b_2 < \dots < b_s$, $c_1 \leq c_2 \leq \dots \leq c_m$.

Define $h'_1: V_1 \rightarrow H^*(E)$ as follows:

$$h'_1(()) = 1,$$

$$h'_1((2k)) = w_{4k}^2 + w_{4k+1}w_{4k-1},$$

$$h'_1((2k+1, 2k+1)) = w_{4k+2}^4,$$

$$\begin{aligned} h'_1((2k+1, 2j+1)) = & w_{4k+2}^2 w_{4j+2}^2 + w_{4k+2}^2 w_{4j+3} w_{4j+1} \\ & + w_{4k+2} w_{4k+1} w_{4j+5} w_{4j} + w_{4k+2} w_{4k+1} w_{4j+5} w_{4j} + w_{4k+2} w_{4k} w_{4j+5} w_{4j+1} \\ & + w_{4k+2} w_{4k+1} w_{4j+3} w_{4j+2} + w_{4k+2} w_{4k} w_{4j+3}^2 + w_{4k+5} w_{4k+1} w_{4j+2} w_{4j} \\ & + w_{4k+5} w_{4k} w_{4j+3} w_{4j} + w_{4k+5} w_{4k} w_{4j+2} w_{4j+2} w_{4j+1} + w_{4k+3}^2 w_{4j+2} w_{4j} \\ & + w_{4k+3} w_{4k+2} w_{4j+3} w_{4j} + w_{4k+3} w_{4k+2} w_{4l+2} w_{4l+1} + w_{4k+3} w_{4k} w_{4j+5} w_{4j} \\ & + w_{4k+3} w_{4k+1} w_{4j+3} w_{4j+1} + w_{4k+3} w_{4k+1} w_{4k+2}^2 + w_{4k+3} w_{4k} w_{4j+3} w_{4j+2}. \end{aligned}$$

$$\begin{aligned} h'_1((a_1, \dots, a_r, b_1, \dots, b_s, c_1, c_1, \dots, c_m, c_m)) \\ = \left(\prod_{i=1}^r h'(a_i) \right) \left(\prod_{i=1}^{s/2} h'((b_{2i-1}, b_{2i})) \right) \prod_{i=1}^m h'((c_i, c_i)). \end{aligned}$$

Note that s must be even, since $\sum a_i + \sum b_i + 2\sum c_i$ is even. Define $h'_2: V_2 \rightarrow H^*(E)$ by

$$h'_2((2k+1)) = w_{4k+2}^2 + w_{4k+3}w_{4k+1},$$

$$h'_2((a_1, \dots, a_r, b_1, \dots, b_s, c_1, c_1, \dots, c_m, c_m))$$

$$= h'_2(a_1, \dots, a_r, b_1, \dots, b_{s-1}, c_1, c_1, \dots, c_m, c_m) \cdot h_2((b_s)).$$

Here s must be odd. Define $h: M \rightarrow H^*(ME)$ by

$$h(x_0 \otimes I) = \Phi h'(I) \quad n(I) \text{ even,}$$

$$h(x_1 \otimes I) = \Phi(h'(I) \cdot (\alpha + \beta)) \quad n(I) \text{ even,}$$

$$h(y_0 \otimes I) = \Phi(h'(I)) \quad n(I) \text{ odd,}$$

$$h(y_1 \otimes I) = \Phi(h'(I) \cdot (\alpha + \beta)) \quad n(I) \text{ odd.}$$

Extend h to all of M by the $\mathcal{A}$ module structure. It is then a tedious but straightforward computation to show that h is well defined. It is easy to show that $h_*: H(M, Q_i) \rightarrow H(H^*(ME), Q_i)$ is a monomorphism, since $h'((i_1, \dots, i_k)) = w_{i_1}^2 w_{i_2}^2 \dots w_{i_k}^2 + \text{correction terms}$. Finally, a counting argument gives that h_* is an isomorphism. Thus by Lemma 3.1, there are free modules F_1 and F_2 and an $\mathcal{A}$ module isomorphism $f: M \oplus F_1 \rightarrow H^*(ME) \oplus F_2$, with $f|_M = g$. Since g is 1-1, as may easily be checked

from the definition, f and F_1 may be chosen so that $F_2 = 0$. Thus f gives an isomorphism from $M \oplus F_1$ and $H^*(ME)$ and the $\mathcal{A}$ module structure of $H^*(ME)$ is determined.

4. Computation of Ω^E

Recall the Adams spectral sequence [1] has E_2 term $E_2^{s,t} = \text{Ext}_{\mathcal{A}}^{s,t}(H^*(X), \mathbf{Z}_2)$ and converges to the 2-primary part of $\pi_*(X)$. So to compute $\Omega_*^E \approx \pi_*(ME)$, it is necessary to compute $\text{Ext}_{\mathcal{A}}^{s,t}(ME, \mathbf{Z}_2)$. But from Theorem 3.6 we have $\text{Ext}_{\mathcal{A}}^{s,t}(H^*(ME), \mathbf{Z}_2) \approx \text{Ext}_{\mathcal{A}}^{s,t}(M_1 \otimes V_1, \mathbf{Z}_2) \oplus \text{Ext}_{\mathcal{A}}^{s,t}(M_2 \otimes sV_2) \oplus \text{Ext}_{\mathcal{A}}^{s,t}(F, \mathbf{Z}_2)$. Since $\text{Ext}_{\mathcal{A}}^{s,t}(F, \mathbf{Z}_2) = 0$ if $s > 0$, consider initially the first two summands. Now $\text{Ext}_{\mathcal{A}}^{s,t}(M_1 \otimes V_1, \mathbf{Z}_2) \approx \text{Ext}_{\mathcal{A}}^{s,t}(M_1, \mathbf{Z}_2) \otimes V_1 \approx \text{Ext}_{\mathcal{A}_1}^{s,t}(M_1^1, \mathbf{Z}_2) \otimes V_1$, and similarly $\text{Ext}_{\mathcal{A}}^{s,t}(M_2 \otimes sV_2) \approx \text{Ext}_{\mathcal{A}_1}^{s,t}(M_2^1, \mathbf{Z}_2) \otimes sV_2$. Since $\mathcal{A}_1$ has only six elements, these are very easy to compute.

LEMMA 4.1.

$$\begin{aligned}\text{Ext}_{\mathcal{A}_1}^{s,t}(M_1, \mathbf{Z}_2) &\approx \text{Ext}_{\mathcal{A}_1}^{s+1, t+1}(\mathbf{Z}_2, \mathbf{Z}_2), \\ \text{Ext}_{\mathcal{A}_1}^{s,t}(M_2, \mathbf{Z}_2) &\approx \text{Ext}_{\mathcal{A}_1}^{s+3, t+6}(\mathbf{Z}_2, \mathbf{Z}_2) \quad s, t \neq (0, 0), \\ \text{Ext}_{\mathcal{A}_1}^{0,0}(M_2, \mathbf{Z}_2) &= \mathbf{Z}_2.\end{aligned}$$

Note that $\text{Ext}_{\mathcal{A}_1}^{*,*}(\mathbf{Z}_2, \mathbf{Z}_2)$ is the $\mathbf{Z}_2$ algebra generated by h_0, h_1, τ, ω with relations $h_1^3 = 0, h_1\tau = 0, \tau^2 = 0, h_1h_0 = 0$ where the degrees of h_0, h_1, τ , and ω are $(1, 1), (1, 2), (3, 7)$ and $(4, 12)$ respectively. For details see Liulevicius [5]. Note that $\text{Ext}_{\mathcal{A}}^{s,t}(H^*(ME), \mathbf{Z}_2) = 0$ if $t - s \not\equiv 0, 1, 2, 4 \pmod{8}$ and $s > 0$, and furthermore $\text{Ext}_{\mathcal{A}}^{*,*}(H^*(ME), \mathbf{Z}_2)$ for $t - s \equiv 0$ or $4 \pmod{8}$ contains only towers except for $s = 0$.

LEMMA 4.2. *The Adams spectral sequence for $\pi_*(ME)$ collapses, i.e., $E_2 = E_\infty$, and all group extensions are trivial.*

Proof. For dimensional reasons, and the fact that nothing except a tower can kill a tower, the only possible non-zero differentials must come from an element of the form $a \otimes \nu$ where $\nu \in V_2$, and $a \in \text{Ext}_{\mathcal{A}}^{0,0}(M_2, \mathbf{Z}_2)$. But the image of this differential would have to be in a dimension congruent to 2 mod 8. Using the $\text{Ext}_{\mathcal{A}}^{*,*}(H^*(M\text{Spin}), \mathbf{Z}_2)$ module structure of $\text{Ext}_{\mathcal{A}}^{*,*}(H^*(ME), \mathbf{Z}_2)$ one sees that the element $\omega \in \text{Ext}_{\mathcal{A}}^{4,12}(H^*(M\text{Spin}), \mathbf{Z}_2)$ has a non-zero product with elements of filtration greater than 0, but $\omega(a \otimes \nu) = 0$. Hence $a \otimes \nu$ must be an ∞ -cycle.

Since the elements of $\text{Ext}_{\mathcal{A}}(F, \mathbf{Z}_2)$ cannot support any non-trivial group extensions [6], and multiples of h_1 have order 2, the only possible group extensions are in dimensions congruent to 1 mod 8, a $\mathbf{Z}_4$ with generator $x_1 \otimes \nu, \nu \in V_1, \nu \neq ()$. Looking at the Ω_*^{Spin} module structure of

Ω_*^E , one has $[x_1 \otimes \nu] = M_\nu \times [x_1]$, where M_ν is the spin manifold corresponding to the sequence ν . Since $[x_1]$ has order 2, so does $[x_1 \otimes \nu]$. Hence all group extensions are trivial, and Ω_*^E has only 2-torsion.

To complete the picture, one must investigate F_2 . Since $H^*(E) \approx H^*(B\text{Spin}) \otimes Z_2(\alpha, \beta)$ as graded groups, it is straightforward to determine the free module summand of $H^*(ME)$ in terms of the summands of $H^*(M\text{Spin})$.

THEOREM 4.3. (a) *For each free module summand F_i in $H^*(M\text{Spin})$ with generator in dimension $n(i)$ there are $k+1$ free summands in $H^*(ME)$ with generators in dimension $n(i)+k$.*

(b) *For each summand of the form $\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1$ in $H^*(M\text{Spin})$ with generator in dimension n , the number of free summands of $H^*(ME)$ with generator in dimension $n+k$ is*

- (i) 0 if k is even,
- (ii) $l+1$ if $k=4l+1$,
- (iii) $l+2$ if $k=4l+3$.

(c) *For each summand of the form $\mathcal{A}/\mathcal{A}Sq^3$ in $H^*(M\text{Spin})$ with generator in dimension n , the number of free summands of $H^*(ME)$ with generator in dimension $n+k$ is*

- (i) 1 if $k=0$,
- (ii) 1 if $k=1$,
- (iii) $3l+2$ if $k=4l+1$, $l>0$,
- (iv) $2l+2$ if $k=4l+2$,
- (v) $3l+4$ if $k=4l+3$,
- (vi) $2l+3$ if $k=4l$, $l>0$.

Proof. By the remark following the proof of Theorem 3.5, the Poincaré polynomial $p(H^*(ME))$ of $H^*(ME)$ is $p(H^*(M\text{Spin}))/ (1-t)^2$. From [2] we have the following Poincaré polynomials:

$$p(H^*(M\text{Spin})) = \prod_{n \geq 3, n \neq 2^r+1} (1-t^n)^{-1},$$

$$p(\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1) = (1-t^4)^{-1}(1-t^6)^{-1} \prod_{n=2^r-1, r \geq 3} (1-t^n)^{-1},$$

$$p(\mathcal{A}/\mathcal{A}Sq^3) = (1+t+t^2+t^3+t^4)p(\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1)$$

and

$$p(\mathcal{A}) = \prod_{n=2^r-1} (1-t^n)^{-1}.$$

From the definition one obtains

$$p(M_1) = p(\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1)(1+t+2t^2+t^3+t^4+t^5)$$

and

$$p(M_2) = p(\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1)(1+2t+t^2+3t^3+2t^4+t^5+t^6).$$

Now

$$(p(\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_2)/(1-t)^2 - p(M_1))/p(\mathcal{A}) = t^3(1-t^4)^{-1} + t(1-t^4)^{-1}(1-t^2)^{-1},$$

$$(p(\mathcal{A}/\mathcal{A}Sq^3)/(1-t)^2 - tp(M_2))/p(\mathcal{A}) = (1+t+t^2+3t^3-t^7)(1-t^4)^{-1}(1-t^2)^{-1},$$

and

$$(p(\mathcal{A})/(1-t)^2)/p(\mathcal{A}) = (1-t)^{-2}.$$

These give easily the numbers above.

The next step is to use this information to compute Ω_n^E . One needs the direct sum decomposition of $H^*(M\text{Spin})$ from [2]. If $n \neq 0 \bmod 4$, Ω_n^E is a direct sum of copies of $\mathbf{Z}_2$. If $n \equiv 0 \bmod 4$, there is one integral summand for each summand of $H^*(M\text{Spin})$ of type $\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1$ or $\mathcal{A}/\mathcal{A}Sq^3$ starting in dimension less than or equal to n . If $n \equiv 1$ or $2 \bmod 8$ a $\mathbf{Z}_2$ is added to Ω_n^E for each summand of $H^*(M\text{Spin})$ of type $\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1$ or $\mathcal{A}/\mathcal{A}Sq^3$ starting in dimensions less than n , and if $n \equiv 3 \bmod 8$, there is a $\mathbf{Z}_2$ for each summand of $H^*(M\text{Spin})$ of type $\mathcal{A}/\mathcal{A}Sq^3$ starting in dimension $n-1$. The rest of Ω_n^E comes from the free summand of $H^*(ME)$, and the number of summands can be computed from Theorem 4.3.

As an example, Ω_{19}^E and Ω_{20}^E are computed. In dimensions less than 21,

$$\begin{aligned} H^*(M\text{Spin}) \approx & (\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1) + s^8(\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1) + s^{10}(\mathcal{A}/\mathcal{A}Sq^3) + s^{16}(\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1) \\ & + s^{16}(\mathcal{A}/\mathcal{A}\bar{\mathcal{A}}_1) + s^{18}(\mathcal{A}/\mathcal{A}Sq^3) + 3^{18}(\mathcal{A}/\mathcal{A}Sq^3) + s^{20}\mathcal{A} \\ ([2]). \text{ Thus} \end{aligned}$$

$$\Omega_{19}^E = 0 \oplus 0 \oplus (\mathbf{Z}_2 \oplus \mathbf{Z}_2) \oplus (\mathbf{Z}_2^8 \oplus \mathbf{Z}_2^4 \oplus \mathbf{Z}_2^8 \oplus \mathbf{Z}_2^2 \oplus \mathbf{Z}_2^2 \oplus \mathbf{Z}_2 \oplus \mathbf{Z}_2) = \mathbf{Z}_2^{28},$$

$$\Omega_{20}^E = \mathbf{Z}^7 \oplus 0 \oplus 0 \oplus (0 \oplus 0 \oplus \mathbf{Z}_2^6 + 0 + 0 + \mathbf{Z}_2^2 + \mathbf{Z}_2^2 + \mathbf{Z}_2) = \mathbf{Z}^7 \oplus \mathbf{Z}_2^{11},$$

where the last term in parentheses corresponds to the free summands in order as given by 4.3.

5. Applications to $\Omega_*^{\text{Spin}}(\mathbf{Z}_2)$

The map $B\text{Spin} \rightarrow E$ induces a rational isomorphism $\Omega_*^{\text{Spin}} \rightarrow \Omega_*^E$, and representatives of the image may be chosen with $\bar{\nu}^*(\alpha) = \bar{\nu}^*(\beta) = 0$.

Hence the composition $\Omega_*^{\text{Spin}} \rightarrow \Omega_*^E \xrightarrow{\theta} \Omega_*^{\text{Spin}}(\mathbf{Z}_2) \rightarrow \Omega_*^{\text{Spin}}$ is multiplication by 2. Thus $\theta: \Omega_*^E \rightarrow \Omega_*^{\text{Spin}}(\mathbf{Z}_2)$ is a rational isomorphism.

COROLLARY 5.1. *All torsion in $\Omega_*^{\text{Spin}}(\mathbf{Z}_2)$ is of order 2.*

COROLLARY 5.2. *The forgetful map $\Omega_*^{\text{Spin}}(\mathbb{Z}_2) \rightarrow \Omega_*^{\text{Spin}}$ is a rational isomorphism.*

To compute the 2-torsion in $\Omega_*^{\text{Spin}}(\mathbb{Z}_2)$ characteristic numbers will be needed, both the integral Pontryagin numbers and the E -Stiefel-Whitney numbers. For any element $x \in H^*(E)$, the E -Stiefel-Whitney number $x(M, \bar{\nu})$ is defined as $\langle \bar{\nu}^*(x), [M] \rangle$. Lemma 4.2 implies the following lemma.

LEMMA 5.3. *If $n \not\equiv 1, 2 \pmod{8}$, then an element of $\Omega_n^E(\mathbb{Z}_2)$ is determined by its characteristic numbers.*

Proof. If $n \not\equiv 0, 1, 2, 4 \pmod{8}$, then the mod 2 Hurewicz map is a monomorphism. For $n \equiv 0, 4 \pmod{8}$, if the Pontryagin numbers of a class $x \in \Omega_n^E$ vanish, then x is not in the kernel of the mod 2 Hurewicz map.

THEOREM 5.4. *If $n \not\equiv 1, 2 \pmod{8}$, $x \in \Omega_n^E$ and $(M, \bar{\nu})$ is a representative of x , then for any map $\rho: M \rightarrow E$ with $\bar{\pi}\rho = \bar{\pi}\bar{\nu}: M \rightarrow BSO \times BO_1 \times BO_1$, (M, ρ) is E -cobordant to $(M, \bar{\nu})$, i.e., the cobordism class is independent of the lifting.*

Proof. This follows from Lemma 5.3, since the characteristic numbers depend only on M and $\bar{\pi}\bar{\nu}$, not on the lifting.

THEOREM 5.5. *$\theta: \Omega_n^E \rightarrow \Omega_n^{\text{Spin}}(\mathbb{Z}_2)$ is an isomorphism if $n \not\equiv 1, 2 \pmod{8}$.*

Proof. Let $x \in \Omega_n^E$, $n \not\equiv 1, 2 \pmod{8}$. If $n > 7$, then by [3] Proposition 1 we may choose a representative $(M, \bar{\nu})$ of x such that $\bar{\nu}_*: \pi_i(M) \rightarrow \pi_i(E)$ is a monomorphism for $i = 1, 2, 3$ and M is connected. Let $\alpha_M = \bar{\nu}^*(\alpha)$, $\beta_M = \bar{\nu}^*(\beta)$. Suppose $\theta(x) = 0$. Then there is a spin manifold W with free orientation preserving involution T such that $\partial(T, W) = \theta(M, \bar{\nu})$. Then W/T has boundary M , and W/T has an E -structure $\bar{\nu}_E$ inducing the Spin structure on W . Hence it is sufficient to show that $(M, \bar{\nu}_E|_M)$ is E -cobordant to $(M, \bar{\nu})$.

Suppose $\alpha_M \neq 0$. Then $\pi_1(M) \approx \mathbb{Z}_2$ or $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ and $\pi_2(M) = 0$. If $\pi_1(M) \approx \mathbb{Z}_2 \oplus \mathbb{Z}_2$, then $\beta_M \neq 0$, and $\alpha_M \neq \beta_M$. Thus the map $\alpha_M: M \rightarrow K(\mathbb{Z}_2, 1)$ or the map $\alpha_M \times \beta_M: M \rightarrow K(\mathbb{Z}_2, 1) \times K(\mathbb{Z}_2, 1)$ induce isomorphisms on the homotopy groups through dimension 3, and hence in $\mathbb{Z}_2$ cohomology through dimension 2. Thus $\ker(\cup \alpha_M): H^1(M) \rightarrow H^2(M)$ is 0, and there is a unique choice of element β_M with $\beta_M \cup \alpha_M = w_2(M)$. Thus by Theorem 5.4, up to E -cobordism there is only one choice of E -structure on M , and hence $(M, \bar{\nu})$ is E -cobordant to 0. If $\alpha_M = 0$, consider the manifold $S^1 \times S^{n-1}$ with an E -structure $\bar{\tau}$ such that $\bar{\tau}^*(\alpha) = \iota_1 \otimes 1$ and $\bar{\tau}^*(\beta) = 0$. By Theorem 5.4, $(S^1 \times S^{n-1}, \bar{\tau})$ is E -cobordant to 0,

and $(M, \tilde{\nu})$ can be replaced by $(M + S^1 \times S^{n-1}, \tilde{\nu} + \tilde{\tau})$. This can be modified to $(M', \tilde{\nu}')$ by surgery so that M' is connected, $\tilde{\nu}'^*: M' \rightarrow E$ is a monomorphism for $1 \leq i \leq 3$ and $\tilde{\nu}'^*(\alpha) \neq 0$. Then proceed as above.

For $n \leq 7$, the theorem follows by construction of explicit representatives.

The computation of $\ker(\theta)$ in dimensions $\equiv 1, 2 \pmod 8$ seems difficult. Note, however, that given Spin manifolds $M_J \times \omega^i \times S^1$ or $M_J \times \omega^i \times S^1 \times S^1$ (notation as in [2]) have an E -structure $\tilde{\nu}$ with $\tilde{\nu}^*(\alpha) = 1 \otimes 1 \otimes \iota_1$ or $1 \otimes 1 \otimes 1 \otimes \iota_1$ and $\tilde{\nu}^*(\beta) = 0$. These are of course not in $\ker(\theta)$, since they are not zero under the composition $\Omega_*^E \rightarrow \Omega_*^{\text{Spin}}(\mathbf{Z}_2) \rightarrow \Omega_*^{\text{Spin}}$. Thus one could hope to get a formula involving E -Stiefel-Whitney numbers for elements x with $\theta(x) = 0$ or $\theta(x)$ one of the products above.

To compute $\Omega_n^{\text{Spin}}(\mathbf{Z}_2)$ for $n \not\equiv 1, 2 \pmod 8$ one needs only to compute Ω_n^E as described at the end of Section 4. For $n \equiv 1, 2 \pmod 8$ this will yield an upper bound on the size of $\Omega_n^{\text{Spin}}(\mathbf{Z}_2)$.

To conclude, specific generators for the torsion elements of Ω_n^E and $\Omega_n^{\text{Spin}}(\mathbf{Z}_2)$ in the range $3 \leq n \leq 7$ are constructed. For these values of n , $\Omega_n^E \approx \Omega_n^{\text{Spin}}(\mathbf{Z}_2)$ and the values, computed from the results in Section 4 are:

$$\begin{aligned}\Omega_3^E &= \mathbf{Z}_2 \oplus \mathbf{Z}_2, & \Omega_4^E &= \mathbf{Z}, & \Omega_5^E &= \mathbf{Z}_2 \oplus \mathbf{Z}_2; \\ \Omega_6^E &= 0, & \Omega_7^E &= \mathbf{Z}_2 \oplus \mathbf{Z}_2 \oplus \mathbf{Z}_2.\end{aligned}$$

If $n \equiv 3 \pmod 4$, $\omega_2(\mathbb{R}P^n) = 0$, hence there are two distinct E -structures $\tilde{\nu}_1$ and $\tilde{\nu}_2$ on $\mathbb{R}P^n$, with $\tilde{\nu}_1(\alpha) = i_1$, $\tilde{\nu}_1(\beta) = 0$, $\tilde{\nu}_2(\alpha) = 0$, $\tilde{\nu}_2(\beta) = i$. Thus $\theta(\mathbb{R}P^n, \tilde{\nu}_1) = (S^n, A)$, where A is the antipodal map, and $\theta(\mathbb{R}P^n, \tilde{\nu}_2)$ is the disjoint union of two copies of $\mathbb{R}P^n$ with different Spin structure and involution which switches copies. Similarly $\mathbb{R}P^5$ has an E -structure $\tilde{\nu}$ with $\tilde{\nu}(\alpha) = \tilde{\nu}(\beta) = i$, and $\theta(\mathbb{R}P^5, \tilde{\nu}) = (S^5, A)$. The remaining manifold in dimensions 5 and 7 can be constructed as a bundle over a real projective space with fiber a real projective space. In the notation of Anderson [3], a representative in dimension 5 is $M(2, 3)$ and in dimension 7 is $M(4, 3)$, where $M(i, n)$ is the projective bundle over $\mathbb{R}P^i$ associated to $\lambda \oplus n$, where λ is the Hopf bundle, and n is a trivial bundle of dimension n . From [3], both of these manifolds are orientable, each of these manifolds has an E -structure, and the E -characteristic numbers are different from those of projective space.

REFERENCES

1. J. F. Adams, 'Stable Homotopy Theory,' Springer-Verlag, 1963.
2. D. W. Anderson, E. H. Brown, Jr., and F. P. Peterson, 'Spin Cobordism,' *Bull. Amer. Math. Soc.* 72 (1966), 256-260.

3. P. G. Anderson, 'Cobordism Classes of Squares of Orientable Manifolds,' *Ann. of Math.* (2) 83 (1966), 47-53.
4. R. Lashof, '*Some Theorems of Browder and Novikov on Homotopy Equivalent Manifolds*', University of Chicago.
5. A. Liulevicius, 'Notes on Homotopy of Thom Spectra', *Amer. J. Math.* 86 (1964), 1-16.
6. H. Margolis, 'On the Realizability of Modules over the Steenrod Algebra', *Bull. Amer. Math. Soc.* 78 (1972), 564-567.
7. R. Stong, '*Cobordism Theory*', Princeton Univ. Press, 1968.

The University of Connecticut
Storrs, Connecticut 06268